

Tutorial 04

CS105.3 Database Management Systems

Draw ER diagrams. Make suitable assumptions and use them in showing maximum and minimum cardinality ratio.

Question 01

The ABC Bank has many branches and a large number of customers. A Customer can open different kinds of accounts with the bank. The bank keeps track of a customer by his SSN, name, address, and phone number. Age is used as a factor to check whether he is a major. There are different types of loans, each identified by a loan number. A customer can take out more than one type of loan, and all branches can give loans. Loans have a duration and interest rate. The account holder can enquire about the balance in his account.

Question 02

In this university setup, there are multiple departments, each offering various courses. Professors are pivotal, teaching across different departments and possibly handling multiple courses. Courses are distinguished by unique codes and titles, along with credit values. Some courses have prerequisites, which impact student enrollment. Students are the focal point, enrolling in multiple courses across departments to pursue their academic objectives. Each student has a unique ID and personal information like name and address. The relationships can be summarized: each department offers many courses, each professor teaches many courses, courses can have multiple prerequisites, and each student can enroll in many courses. This arrangement fosters a dynamic environment where departments offer courses, professors instruct, and students pursue their academic paths by selecting courses aligned with their goals.

Question 03

The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations. A department controls a number of projects, each of which has a unique name, a unique number, and a single location. The database will store each employee's name, Social Security number, address, salary, sex (gender), and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. It is required to keep track of the current number of hours per week that an employee works on each project, as well as the direct supervisor of each employee (who is another employee). The database will keep track of the dependents of each employee for insurance purposes, including each dependent's first name, sex, birth date, and relationship to the employee.